**Government of Karnataka**

**Department of Technical Education**

**Board of Technical Examinations**

**Course: Thinking Programming with Python** **Course Code: 25CS21I**

**Time: 3 Hrs** **Max Marks: 100**

***Instructions***: For Part-A questions, only the first written answers will be considered for evaluation.

**PART A**

**Select the correct answer from the choices given 15x1 = 15**

1. What is the purpose of using pseudocode in computational thinking?
   A) To execute a program B) To install software
   C) To plan a solution using simple, readable steps D) To draw complex hardware diagrams
2. What type of problem is "sorting exam scores from highest to lowest"?
   A) Searching problem B) Pattern recognition problem
   C) Sorting problem D) Decomposition problem
3. Which of the following best defines algorithmic thinking?
   A) Writing code without syntax errors
   B) Designing a sequence of steps to solve a problem
   C) Drawing diagrams to represent data
   D) Learning multiple programming languages
4. Which of these is NOT a benefit of using algorithms?
   A) Solving problems systematically B) Repeating known solutions
   C) Reducing problem size D) Making random guesses
5. Which term refers to finding similarities in problems to reuse solutions?
   A) Abstraction B) Pattern Recognition C) Algorithm D) Flowcharting
6. A student designs a step-by-step plan to calculate the average marks of a class. This plan is an example of:
   A) Abstraction B) Algorithm C) Debugging D) Simulation
7. Which operator has the highest precedence?
   A) + (Addition) B) ** (Exponent)
   C) == (Comparison) D) not (Logical NOT)
8. What is a syntax error in programming?
   A) Incorrect program logic
   B) Violation of language rules (e.g., missing colon or parentheses)
   C) A typo in a comment D) Using an undefined variable

9. What is the correct symbol for a single-line comment in Python?

   A) //  B) <!-- -->  C) /* */  D) #

10. Which of the following is NOT a data type in Python?

    A) int  B) float  C) real  D) str

11. Which function is used to take input from the user in Python?

    A) scan()  B) read()  C) input()  D) get()

12. Which of the following is used for string formatting in Python?

    A) %  B) f-strings  C) format()  D) All of the above

13. Which of the following conditions makes a while loop infinite?

    A) while x > 0: with decreasing x  B) while True: with no break

    C) while x != 0: and x = 0  D) while False:

14. How many times will "Hello" print?

```
for i in range(3):
  for j in range(2):
    if j == 1:
      continue
    print("Hello")
```

    A) 2  B) 3  C) 6  D) 1

15. What does the break statement do in a loop?

    A) Skips the next iteration  B) Exits the loop immediately

    C) Repeats the loop  D) Ends the program

**I. Fill in the blanks by choosing appropriate answer from those given in the bracket:**

**(programming, import, algorithm, true, No,)** **5X1 = 05**

1. A recipe is a real-life example of a(n) _____.
2. The process of writing instructions for computers to solve problems is called _____.
3. _____ is the output of: if 1 and 0: print("Yes") else: print("No").
4. A while loop becomes infinite if the condition is _____ and there is no break.
5. To use the math module, you first need to _____ it.

**PART B**

**II. Answer any FIVE questions:** **5X2 = 10**

1. Define algorithmic thinking.
2. Mention any two types of problems where algorithmic solutions are commonly used.
3. Define a variable in programming
4. What is a loop in programming?

5. Name any two tools or techniques used for debugging in Python.
6. What is the difference between compile-time and run-time errors in Python?
7. Define a function in the context of Python programming.
8. What is modular programming?

## PART C

**VI. Answer any FIVE questions:** **5X3 = 15**

1. Explain algorithmic thinking with a tea making example.
2. What is the significance of data types in Python?
3. Differentiate between syntax and semantic errors in Python
4. How are constants and variables different? Give examples.
5. Explain the difference between for loop and while loop in Python with example scenarios.
6. Why is testing important after fixing errors in a program?
7. Differentiate between modular programming using functions vs packages.
8. What are the advantages of using built-in modules in Python? Give examples.

## PART D (Section I)

**III. Answer any FIVE questions:** **5X5 = 25**

1. Describe the four pillars of computational thinking with suitable examples for each.
2. What are tokens in Python? List and explain different types of tokens with examples.
3. Explain the rules for creating valid variable names in Python with examples.
4. Write Python statements to print the expected output using three different string formatting methods given the following data:

   name = "Mark"
   subject = "Thinking Programming"
   marks = 95
   *Expected Output* : Student Mark has scored 95 marks in Thinking Programming.

5. Explain the different types of flow control statements in Python. Describe their purpose and give one example for each:

6. The following program is supposed to skip the number 4 and 14 and print the rest from 1 to 20. Find and fix the errors.

```
for i in range(1,20):
if i = 4:
    continue
print i
```

7. Why are functions needed in Python programming? Explain how to create and call a function with and without arguments. Give examples.
8. What is the significance of using modules and packages in Python? How do they help in large program development?

**PART D (Section II)**

**IV. Answer any THREE questions: 10X3 = 30**

1. Identify the tokens in the following Python statement and explain each of them with its category

```
while count < 10:
    result = result + value_1 - 3
```

2. Create a program that simulates an ATM machine to check:
   - If the entered PIN is correct, allow the user to proceed.
   - Validate that the withdrawal amount doesn't exceed the account balance.
   - Ensure that the withdrawal amount is in multiples of a specific denomination (e.g., Rs100).
3. Create a Python program for a bookstore that wants to automate discount calculation based on the total purchase amount.

**Discount Policy:**

| Total Purchase Amount (₹) | Discount Offered |
|---|---|
| ≥ ₹5000 | 25% |
| ₹3000 to < ₹5000 | 15% |
| ₹1000 to < ₹3000 | 5% |
| < ₹1000 | No Discount |

Follow best programming practices (variable naming, indentation, and comments).

4. The following Python code calculates the average marks of a student in 3 subjects and prints the corresponding grade:

```
math = float(input("Enter marks in Mathematics: "))
science = float(input("Enter marks in Science: "))
english = float(input("Enter marks in English: "))
average = (math + science + english) / 3
if average >= 90:
   grade = "A"
elif average >= 75:
   grade = "B"
elif average >= 50:
   grade = "C"
else:
   grade = "F"
print(f"Average Marks: {average:.2f}")
          print(f"Grade: {grade}")
```

Rewrite the above program using a modular approach by defining and using appropriate functions such as:

- A function to get input from the user
- A function to calculate average
- A function to determine grade
- A main function to organize the program flow

5. Describe the different types of function arguments in Python with suitable examples.

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